



PHINMA University of Iloilo
College of Information Technology Education

ProfitLens

A Web-Based Platform for Smart Profit Monitoring and Business Insights

for Auto Parts Sales

System Documentation

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Part I

Project Overview

Chapter 1

Introduction

This chapter situates ProfitLens within the context of small and medium enterprise (SME) financial management, articulates the specific operational problems that motivated its creation, states the study's objectives, defines its functional boundaries, explains who stands to benefit from the platform, and provides precise working definitions of the technical vocabulary used throughout this document.

1.1 Background of the Study

In the Philippine retail landscape, small and medium enterprises operating within the auto parts sector occupy a critical but often under-served niche. These establishments serve mechanics, fleet operators, and individual vehicle owners who depend on timely access to engine components, consumables such as oils and filters, and mechanical hardware including bearings and belts. Despite the commercial importance of these businesses, many of them — particularly those operating at the scale of a single-branch owner-managed store — continue to rely on manual financial record-keeping methods that date back several decades.

The particular context from which ProfitLens originates is a local auto parts store owned and operated by French Joy Salvador in Cebu City. Observations gathered during the requirements-gathering phase of this project revealed that the store's financial processes followed a pattern common to many similarly scaled Philippine SMEs: daily sales were tallied on paper ledgers or basic spreadsheets maintained without formulas, expenses were tracked through a combination of physical receipts and informal notes, and monthly profit was computed retrospectively by manual subtraction — a process that consumed significant staff time and regularly produced inconsistent summaries.

Several structural inefficiencies were identified through direct observation and stakeholder interviews. First, the calculation of monthly net profit required manually aggregating tens or hundreds of individual sales entries against a correspondingly informal list of operating expenses. This process was not only time-consuming but inherently prone to arithmetic errors, missing entries, and duplicated records. Second, the absence of a centralized data repository meant that historical financial performance — revenue trends, seasonal variation in demand for specific product categories, or month-over-month growth — was effectively inaccessible. No mechanism existed for generating the kind of comparative analysis that might allow the owner to identify which months were consistently strong or which product lines commanded the highest margins. Third, operating expenses regularly exceeded available budget allocations without triggering any alert or warning, a condition that the owner only discovered after the fact during periodic manual reconciliation.

These challenges are consistent with broader patterns documented in the SME financial management literature. Research consistently demonstrates that small businesses that lack automated, centralized financial monitoring systems are more likely to experience cash flow difficulties, less likely to make evidence-based investment decisions, and more vulnerable to the compounding effects of unnoticed expenditure creep. In the Philippine context specifically, where many SMEs operate on thin margins and have limited access to formal financial advisory services, the consequences of inadequate financial visibility can be severe.

ProfitLens was conceived and built as a direct, targeted response to these conditions. Rather than attempting to replicate the full feature set of enterprise-grade accounting software — which would impose an unnecessary learning curve on store-level staff and require ongoing technical support that is practically unavailable to most small Philippine businesses — the platform is deliberately scoped to address the specific, concrete problems identified at the Salvador store. It provides automated computation of net profit from sales and expense entries, centralized storage of all financial data in a normalized MySQL database, real-time visual presentation of revenue trends through bar charts and line graphs covering the full January-to-December calendar year, and a proactive expense monitoring system with category-based breakdowns that visually highlight high-spend areas relative to revenue.

The platform is delivered as a web-based application built on a PHP-MySQL-HTML-CSS-JavaScript technology stack. This combination of technologies was chosen deliberately for its accessibility and maintainability within the Philippine educational and small-business context: PHP and MySQL are widely understood by local developers, require no commercial licensing fees, and are fully supported by the XAMPP local server environment commonly used in academic and SME settings. The web-based delivery model means that the system is accessible from any device with a browser, eliminating the need for platform-specific client software installation.

Beyond its immediate operational utility, ProfitLens is designed to serve as a proof-of-concept and reference model for future automated financial monitoring systems targeting Philippine SMEs. Its architecture is deliberately documented in sufficient detail — as reflected in this document — to allow other development teams, student researchers, or business owners to understand, replicate, adapt, or extend the system for their own contexts.

1.2 Statement of the Problem

The central operational challenge addressed by ProfitLens can be summarized as follows: the auto parts store operated by French Joy Salvador in Cebu City lacked any automated system for tracking, aggregating, visualizing, or reporting financial data. All financial management was conducted through manual methods that were inadequate in precision, transparency, and actionability.

The following four specific problems, identified through direct observation of the store's operations and through structured interviews with the owner and staff, define the problem space that ProfitLens was engineered to resolve:

Problem 1 — Profit Monitoring Difficulty

Calculating monthly net profit required the manual aggregation of all individual sales transactions across the month, followed by a manual subtraction of all recorded expenses. This process was performed at the end of each month, typically by the owner herself, and regularly required several hours of work to complete. The manual nature of the calculation introduced multiple points of potential error: individual entries could be missed, transcription errors could propagate through the calculation, and the lack of any validation mechanism

meant that erroneous totals could be accepted without detection. The result was a monthly profit figure that the owner acknowledged she could not rely upon with full confidence, and that provided no granular insight into the relative contribution of different products or expense categories to the overall financial outcome.

ProfitLens addresses this problem by automatically computing net profit in real time whenever sales and expense records are entered into the system. The computation is performed using a clearly defined formula applied consistently by the PHP backend, ensuring mathematical accuracy and eliminating the manual aggregation step entirely.

Problem 2 — Absence of Structured Financial Reports

No formal mechanism existed for producing monthly, quarterly, or annual financial summaries. Historical financial data was stored in physical documents and informal spreadsheets that were not organized in a way that would support retrospective reporting. When the owner needed to review past performance — for example, to assess whether a particular month's revenue was sufficient to sustain a planned purchase of additional inventory — she was required to manually reconstruct the relevant figures from archived receipts and notes. This process was impractical for any period longer than a few months and impossible for periods where physical records had been lost or damaged.

ProfitLens resolves this by maintaining all financial data in a structured MySQL database and providing a dedicated Financial Reports module that generates comprehensive profit-and-loss summaries, monthly expense breakdowns, and a full January-to-December grouped bar chart comparing revenue against expenses — all available on demand within the system, with no manual reconstruction required.

Problem 3 — No Trend Analysis Capability

Without a centralized data store and automated reporting tools, it was impossible for the store owner to visualize performance trends, identify seasonal demand patterns, compare monthly or annual results, or make data-driven predictions about future revenue. The store had no way of knowing, for instance, whether June was consistently its strongest month, whether demand for oil-based consumables peaked before the rainy season, or whether year-over-year revenue was growing, declining, or stable. These are the kinds of insights that enable proactive business decision-making — adjusting inventory orders in advance of high-demand periods, negotiating better purchase terms with suppliers during low-revenue months, or setting realistic revenue targets — and their absence left the owner operating largely on intuition rather than evidence.

ProfitLens addresses this through its Revenue Analysis module, which presents monthly revenue trends in bar and line chart formats, supports year-over-year comparison across any two calendar years, and identifies the top five products by revenue and units sold — all computed dynamically from the database and rendered visually in the browser without requiring any external chart library.

Problem 4 — Undetected Overspending

Operating expenses at the store encompassed a diverse range of cost categories: utilities, office supplies, advertising, salaries and wages, and various miscellaneous operational costs. Without automated monitoring, the cumulative total of these expenses frequently exceeded the budget available for a given month without the

owner becoming aware of the overrun until she conducted her end-of-month manual reconciliation. This lag between the occurrence of overspending and its detection meant that corrective action — such as deferring a discretionary expense or renegotiating a supplier payment schedule — could not be taken in time to prevent the financial impact from materializing.

ProfitLens addresses this through its Expense Tracking module, which maintains a real-time running total of all recorded expenses for the current month, categorizes them for granular visibility, and provides visual indicators that highlight high-spend categories relative to overall revenue — giving the owner and staff an always-current view of the expense position without requiring any manual computation.

1.3 Objectives of the Study

1.3.1 General Objective

The overarching objective of this study is to design, develop, and deploy ProfitLens — a smart, web-based profit monitoring platform — that delivers automated business insights, accurate financial tracking, and proactive alerts for overspending in an auto parts sales and services environment. The platform is intended to replace the manual financial management practices currently in use at the reference business and to provide its users with a reliable, easy-to-use, and visually informative system that supports daily operational decision-making and longer-term financial planning.

1.3.2 Specific Objectives

In support of the general objective, this study aims to accomplish the following four specific objectives:

- 1. To provide a module for tracking monthly profit** by recording individual sales transactions — each defined by a selected product, an entered quantity, and a specified sale date — against a live-updated register of product costs and operating expenses. The module must automatically compute the total value of each sale from the product's selling price, deduct the sold quantity from the corresponding stock record, and present a continuously updated net profit figure computed as the difference between the period's total sales revenue and total recorded expenses.
- 2. To implement a financial reporting module** that displays monthly, quarterly, and annual profit-and-loss summaries within the system's web interface. The module must present a full January-to-December comparison of revenue and expenses in both tabular and graphical formats, support the selection of specific calendar years for historical review, and make all reported data available without requiring the export of raw records to an external application for processing.
- 3. To implement automated revenue insights** that display bar graphs comparing monthly revenue versus expenses across the full calendar year (January through December), supplemented by line chart representations of revenue trends and year-over-year comparative visualizations. These visual outputs must be computed dynamically from the database without relying on any external charting library, ensuring that the system remains self-contained and does not depend on third-party service availability.

4. To develop an expense tracking module with category-based expense breakdowns — covering categories including Office Supplies, Advertising, Utilities, and Salaries and Wages — and visual indicators that highlight high-spend areas relative to total revenue for the current period. The module must support the addition, editing, and deletion of individual expense records, maintain accurate running totals, and provide the owner and staff with immediate visual feedback when expense levels in any category represent a disproportionately high share of the period's revenue.

1.4 Scope and Delimitation

1.4.1 Functional Scope

ProfitLens covers the following functional areas, all of which are fully implemented in the delivered system:

- **User Authentication:** Secure login and registration for admin-level users, including password hashing using PHP's `password_hash()` function and session-based access management.
- **Product Management:** The ability to add, edit, and delete products with associated attributes including name, category, selling price, cost price, and current stock quantity.
- **Sales Recording:** A form-based interface for entering sales transactions, with automatic total calculation, stock deduction, and validation to prevent recording of sales that would result in negative stock.
- **Expense Tracking:** A categorized expense logging system supporting addition, editing, and deletion of records, with real-time totaling and category-level breakdowns.
- **Revenue Analysis:** A dedicated module presenting monthly revenue trends in bar and line chart formats, year-over-year comparisons, and a ranked list of top-performing products.
- **Profit and Loss Reporting:** A comprehensive reporting module providing a January-to-December grouped bar chart of revenue versus expenses, a tabular profit-and-loss summary, and a monthly expense breakdown report with year selection capability.
- **Financial Dashboard:** A centralized landing page presenting key performance indicators — net profit for the current month, growth percentage relative to the prior month, a mini bar chart of the current year's revenue distribution, the best-selling product — alongside visual summaries from the expense and revenue modules.
- **Role-Based Access Control:** Differentiated access levels for admin users (full system access) and standard users (restricted access), enforced through PHP session validation on every protected page.

1.4.2 Delimitations

The following functional areas are explicitly outside the scope of ProfitLens as designed and implemented. These delimitations reflect deliberate decisions made to keep the system focused, maintainable, and appropriate for the scale and technical environment of the reference business:

- **Point-of-Sale Hardware Integration:** ProfitLens does not interface with barcode scanners, receipt printers, cash drawers, or any other physical POS hardware. Sales are entered manually through the web interface.

- **Payroll Processing:** While salaries and wages are trackable as an expense category, ProfitLens does not compute individual employee payroll, manage attendance records, or generate payslips.
 - **Tax Filing and Compliance:** The system does not generate BIR-compliant tax documents, compute VAT obligations, or integrate with any government tax reporting platform.
 - **External Accounting System Integration:** ProfitLens does not connect to or synchronize data with any third-party accounting application such as QuickBooks, Xero, or any cloud-based ERP system.
 - **Multi-Branch Management:** The system is designed for a single-location business. It does not support consolidated reporting across multiple branch locations or inter-branch inventory transfer tracking.
 - **Customer Relationship Management (CRM):** ProfitLens does not maintain a customer database, track individual customer purchase histories, or support customer communication functions.
 - **Full Warehouse Management:** While product stock quantities are maintained and updated upon each sale, the system does not support purchase order management, supplier tracking, reorder-level alerts, or warehouse location tracking.
 - **Email Notifications:** Although a forgot-password page is provided, the system does not send automated email notifications of any kind in its current implementation, as email server configuration is outside the deployment environment's scope.
-

1.5 Significance of the Study

The development and documentation of ProfitLens carries significance that extends beyond the immediate operational benefits to the reference business. The following stakeholder groups stand to benefit from the outcomes of this study:

1.5.1 Business Owners

For the primary user of the system — the small auto parts store owner — ProfitLens provides real-time profit visibility that was previously unavailable without laborious manual calculation. The platform's dashboard presents net profit for the current month alongside a growth indicator showing whether performance is improving or declining relative to the previous period. The full January-to-December revenue versus expense bar chart enables the owner to identify seasonal patterns, assess whether the business is on track relative to prior years, and make informed decisions about inventory investment, staffing levels, and discretionary spending. The categorized expense breakdown gives immediate visibility into which cost categories are consuming the largest proportions of revenue, enabling targeted cost management rather than undifferentiated expense reduction.

1.5.2 Staff Members

Store staff benefit from a guided, form-based system for recording sales and expenses that replaces the informal and inconsistency-prone manual process. The system's automatic total calculation for each sales entry removes the arithmetic burden from staff and eliminates the most common source of data entry errors. The searchable product dropdown in the sales recording form reduces the time required to locate and select the

correct product, particularly as the product catalog grows. Staff can complete routine financial recording tasks more quickly, with greater confidence in the accuracy of the records they are creating.

1.5.3 Researchers and Future Developers

ProfitLens serves as a practical reference model for researchers interested in automated financial monitoring systems designed for small businesses in the Philippine context. The system's architecture — a PHP-MySQL client-server application with inline SVG chart generation, role-based access control implemented through PHP sessions, and a deliberately limited feature scope targeted at a specific operational problem — illustrates design decisions that are transferable to analogous contexts. Future researchers can use the system as a baseline against which more feature-rich or differently architected solutions can be compared, and can build upon the theoretical framework and literature review provided in Chapter 2 as a foundation for further scholarly inquiry.

1.5.4 The Auto Parts Sales and Services Industry

More broadly, ProfitLens demonstrates the feasibility of developing cost-effective, fit-for-purpose financial monitoring solutions for small auto parts retailers in the Philippine market. By standardizing the data structures and reporting formats used for financial tracking within a single store, the platform establishes a basis upon which industry-level data collection and benchmarking could eventually be built. If similar systems were adopted across multiple stores — whether through replication of the ProfitLens model or through adaptation to different technology stacks — it would become possible to develop aggregate insights about the financial performance, cost structures, and demand patterns of the Philippine auto parts retail sector that are currently unavailable due to the fragmented and informal nature of financial record-keeping across the industry.

1.5.5 Academic Institutions

For BSIT students and faculty at PHINMA University of Iloilo and comparable institutions, ProfitLens provides a thoroughly documented capstone project that demonstrates the integration of multiple competencies developed across the BSIT curriculum: user-centered interface design, relational database design and implementation, PHP backend development with security best practices, agile project management methodology, and the application of systems thinking to a real-world business problem. The platform's code, this documentation, and the theoretical framework presented in Chapter 2 collectively constitute a resource that future capstone teams can consult as they undertake their own projects.

1.6 Definition of Terms

The following terms are used throughout this documentation with the specific meanings defined below. Where terms have both a general meaning and a system-specific technical meaning, the system-specific meaning is provided.

Term	Definition as Used in This Document
ProfitLens	The web-based platform for smart profit monitoring and business insights that is the subject of this documentation. Built with PHP 8, MySQL, HTML5, CSS3, and JavaScript, and deployed on an Apache/XAMPP local server environment.
Dashboard	The system's primary landing page after login (<code>dashboard.php</code>), which displays real-time financial summaries including net profit for the current month, growth percentage relative to the prior month, the best-selling product, expense breakdown, and revenue trend visualizations.
Sales Module	The system module implemented in <code>profit.php</code> that enables admin users to record sales transactions by selecting a product from a searchable dropdown, entering a sale quantity, and specifying a sale date. The module automatically computes the transaction total from the product's stored selling price and deducts the sold quantity from the product's stock record.
Expense Tracking	The system module implemented in <code>expenses.php</code> that provides functionality for adding, editing, and deleting business expense records, categorized by type (Office Supplies, Advertising, Utilities, Salaries and Wages, and others). The module maintains real-time monthly and annual expense totals.
Revenue Analysis	The system module implemented in <code>revenue.php</code> that presents visual representations of monthly revenue trends, year-over-year performance comparisons, and a ranked listing of top-performing products by units sold and total revenue.
Financial Reports	The system module implemented in <code>reports.php</code> that presents a comprehensive January-to-December grouped bar chart comparing revenue and expenses, a tabular profit-and-loss summary, and a detailed monthly expense breakdown report. Supports year selection for historical review.
Product Management	The system module implemented in <code>products.php</code> that provides full create, read, update, and delete (CRUD) functionality for product records, each of which carries attributes including name, category, selling price, cost price, and current stock quantity.

Authentication	The process by which the system verifies a user's identity before permitting access to protected pages. ProfitLens implements authentication through a login form (<code>index.php</code>) that validates submitted credentials against hashed passwords stored in the <code>users</code> database table using PHP's <code>password_verify()</code> function.
Session	PHP's built-in mechanism for maintaining state across multiple HTTP requests from the same user, implemented via the <code>\$_SESSION</code> superglobal. ProfitLens stores the authenticated user's <code>user_id</code> , <code>email</code> , and <code>role</code> in the session upon successful login, and checks the session on every protected page to verify continued authorization.
Net Profit	The financial metric computed by the system as: <i>Net Profit = Sum of Sales Totals – Sum of Expense Amounts</i> for a given period. Displayed on the dashboard, profit module, and reports module. A positive value indicates a profitable period; a negative value indicates that expenses exceeded revenue.
formatMoney()	A PHP helper function defined in <code>includes/config.php</code> that formats a numeric amount as a Philippine Peso currency string. Syntax: <code>formatMoney(\$amount)</code> returns a string in the form <code>₱15,000.00</code> . Used consistently across all modules wherever monetary amounts are displayed to the user.
Prepared Statement	A database query technique in which SQL commands are compiled separately from user-supplied data values, which are then bound to the compiled query as parameters. ProfitLens uses MySQL's prepared statement implementation (via PHP's <code>mysqli</code> extension) throughout the system to prevent SQL injection attacks.
Admin Role	The user role designation (<code>role = 'admin'</code> in the <code>users</code> table) that grants full access to all system modules, including the ability to add, edit, and delete products, sales, and expenses. Non-admin users who attempt to access admin-protected pages are redirected to an access-denied page by the <code>requireAdmin()</code> function.
XAMPP	A cross-platform local web server environment bundling Apache (web server), MariaDB/MySQL (database server), and PHP. ProfitLens was developed and tested on XAMPP and is designed for deployment in XAMPP or any equivalent Apache/PHP/MySQL environment.
profit_lens Database	The MySQL database that stores all system data. Contains four primary tables: <code>users</code> , <code>products</code> , <code>sales</code> , and <code>expenses</code> . All system modules interact with this database

	exclusively through the <code>getDB()</code> connection function defined in <code>includes/config.php</code> .
SME	Small and Medium Enterprise. In the Philippine context and as used in this document, refers to businesses with fewer than 200 employees and annual revenue below the threshold defined by the Philippine Department of Trade and Industry. The auto parts store that serves as the reference business for ProfitLens is classified as a micro-enterprise by this definition.
DFD	Data Flow Diagram. A graphical representation of how data moves through a system, showing sources, destinations, processes, and data stores. ProfitLens is documented using DFD Level 0 (context diagram showing high-level interactions) and DFD Level 1 (detailed process diagram) conventions.
ERD	Entity Relationship Diagram. A visual representation of the relationships between the entities (tables) in a database. The ProfitLens ERD defines the structure of and relationships among the <code>users</code> , <code>products</code> , <code>sales</code> , and <code>expenses</code> tables.

Note on Currency Formatting: All monetary amounts presented in system interfaces and in this documentation are expressed in Philippine Pesos (PHP), formatted as

using the

helper function. This formatting is applied consistently across all modules and reports to ensure that all financial figures are immediately recognizable as peso-denominated values.

With the conceptual and definitional groundwork established in this chapter, the following chapter reviews the theoretical and empirical literature that informed the design of ProfitLens, situating the system within the broader scholarly conversation about financial automation, SME digital transformation, and the specific challenges of the Philippine small business context.